

Juan Pablo Roldan

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PROFILE

PhD student in aerospace engineering specializing in hypersonic and multiphase flows. Develops numerical methods for shock-driven droplet breakup and high-speed multiphase transport, with application to hypersonic rain environments, entry aerothermodynamics, and liquid-fueled propulsion. Complementary focus on GPU parallelization, reduced-order and data-driven modeling, and scientific software development for large-scale simulation. Seeking to pair computational work with experimental validation in high-speed flow facilities.

EDUCATION

- PhD Aerospace Engineering, Aerodynamics and Propulsion** Aug 2026 – IP
Embry-Riddle Aeronautical University (ERAU), Daytona Beach, FL, USA
- MS Aerospace Engineering, Aerodynamics and Propulsion** (Dual Degree, GPA 4.00) Aug 2024 – Aug 2026
Embry-Riddle Aeronautical University (ERAU), Daytona Beach, FL, USA
Universidad del Valle (UV), Cali, Colombia
- BS Aerospace Engineering, Minor in Computer Science** (GPA 4.00) Aug 2021 – May 2024
University of Central Florida (UCF), Orlando, FL, USA

CONFERENCE PAPERS

- Juan P. Roldan**, Brendon A. Cavainolo, Sheryl M. Grace, Michael P. Kinzel, Guillermo A. Jaramillo, “Initial Perturbation Effects on Surface Instability Wavelength Selection in High-Mach Drop Aerobreakup,” *AIAA SciTech*, Jan 2027. (Abstract submitted)
- Juan P. Roldan**, Sutharsan Satcunanathan, Georgios Goinis, “POD Airfoil Dimensionality Reduction for Compressor Cascade Optimization,” *AIAA SciTech*, Jan 2027. (Abstract submitted)
- Juan P. Roldan**, Brendon A. Cavainolo, Rebecca C. Shannon, Sheryl M. Grace, Michael P. Kinzel, “Effect of Entrapped Non-Condensable Gas on Shock-Driven Droplet Deformation: Implications for Hypersonic Rain Environments,” *AIAA International Space Planes and Hypersonic Systems and Technologies Conference*, Jul 2026.
- Eric M. Yoerg, **Juan P. Roldan**, Donald Tuten, Michael Potash, Rui Harrison, Cassandra J. Durst, Michael P. Kinzel, Surabhi Singh, “Development of an Exploding Wire Facility to Study Shock Propagation at Embry-Riddle Aeronautical University,” *AIAA SciTech*, Jan 2025.

INDUSTRY-PARTNERED RESEARCH DEVELOPMENT

- Modernization of CFD Tools for Advanced Propulsion Applications (Not Awarded)** Submitted May 2026
NASA STTR Phase I | Partners: Advanced CFD Solutions, Worcester Polytechnic Institute
- Led technical development of the proposal concept, centered on multiphase-flow modeling for liquid-fueled detonation and shock-droplet interaction applications.
- Augmenting Classical HPC with Quantum Computing Capabilities for CFD (Not Awarded)** Submitted Sep 2025
DoD DTRA STTR Phase I | Partners: Advanced CFD Solutions, ERAU; with support from Oak Ridge National Laboratory
- Led technical development of the proposal concept, targeting faster computational air-blast prediction by offloading portions of the CFD workload from classical HPC to quantum hardware.

RESEARCH EXPERIENCE

Hypersonic and High-Speed Flows

- Numerical Study of Shock-Induced Droplet Deformation with Cavitation Nuclei Seeding** Aug 2024 – IP
Graduate Researcher, ERAU Computational Fluids and Aerodynamics Laboratory (M. P. Kinzel), and Universidad del Valle, IMPETUS INDOMITUS (G. A. Jaramillo)
- Simulating shock-driven droplet deformation and aerobreakup using the Multi-Component Flow Code (MFC), resolving bubble dynamics from seeded cavitation nuclei and entrapped non-condensable gas.
 - Quantifying how initial perturbations govern surface instability wavelength selection during high-Mach aerobreakup, with implications for hypersonic rain erosion environments.
 - Developing linear stability analysis capability and evaluating paths to incorporate data-driven and reduced-order methods.
- Enhanced Backshell Heating Estimates for Dragonfly's Aeroshell** Jun 2023 – Aug 2023
Visiting Researcher, NASA Ames Research Center, Aerothermodynamics Branch | NASA MUREP-STAR

- Simulated chemically reacting hypersonic flow over the Dragonfly aeroshell during Titan entry using US3D, targeting the separated backshell wake where heating estimates are most grid-sensitive.
- Built and evaluated a metric-based mesh adaptation workflow to improve backshell heating predictions without uniform grid refinement.
- Completed as a 10-week embedded placement at a national research center, working within an operational aerothermodynamics team.

Hypersonic Waverider Design and Analysis Tool

Jan 2025 – May 2025

Lead Developer | github.com/JuanPabloRoldan/HyPyRider

- Led development of a Python-based rapid design and analysis tool for waverider geometries, following the design methodology of K. G. Bowcutt's thesis.

Multiscale Simulations of Scramjet Inlets with Dispersed Particles

May 2024 – Aug 2024

Undergraduate Researcher, UCF Department of Mechanical and Aerospace Engineering (M. P. Kinzel)

- Conducted 2D steady non-reacting simulations of scramjet inlets under nominal flow conditions.
- Captured droplet trajectories and local flow variables via a dispersed multiphase model, supplying initial and boundary conditions for microscale single-droplet simulations.

Reduced-Order Modeling and Spacecraft Applications

POD Dimensionality Reduction for Compressor Cascade Optimization

Jun 2025 – Aug 2025

Visiting Researcher, DLR Cologne, Institute of Propulsion Technology, Fan and Compressor Group | NSF IRES

- Investigated proper orthogonal decomposition for reducing design-space dimensionality in compressor cascade optimization.
- Characterized the tradeoff between dimensionality reduction and loss of optimization accuracy.
- Completed as a 10-week embedded placement at a national aerospace research center abroad.

Plume Impingement Analysis: RPOD Maneuvers to the Lunar Gateway

Sep 2023 – May 2024

Co-Lead Developer | github.com/plume-kit/PyRPOD | Mentor: J. S. Pitt

- Co-led development of a rapid-analysis tool characterizing visiting-vehicle thruster configurations by impingement, fuel usage, and safety metrics.
- Implemented rarefied gas dynamics and gas-surface interaction models.

GRANTS AND AWARDS

NSF International Research Experience for Students — \$13,000

Jun 2025 – Aug 2025

Funded 10 weeks as a visiting researcher at DLR Cologne.

Benjamin A. Gilman International Scholarship — \$6,000

Jan 2024

Funded a short-term study abroad program in Sao Paulo, Brazil.

NASA MUREP Scholarship for Training, Advancement, and Research (MUREP-STAR) — \$10,000

Mar 2023 – Aug 2023

Funded a three-month placement as a visiting researcher at NASA Ames Research Center.

UCF National Hispanic Recognition Program Scholarship — \$48,000

Aug 2021 – May 2024

Funded four years of undergraduate study in aerospace engineering.

SKILLS AND SOFTWARE

CFD Software: Multi-Component Flow Code (MFC), ALPACA (Technical University of Munich), US3D, STAR-CCM+

Programming: Python, C, MATLAB, Java, CUDA

HPC and Tools: Linux, SLURM, PBS, Git/GitHub

ADDITIONAL EXPERIENCE

Academic Exchange, Universidad del Valle, Cali, Colombia

Aug 2025 – May 2026

- Completed coursework and research under the ERAU-UV dual-degree agreement; supported peer fellowship applications and English-language practice, and helped establish collaborative research networks.

Contemporary Challenges in Global Aerospace, UCF and Embraer, Sao Paulo, Brazil

Jan 2024 – May 2024

Faculty-embedded short-term program on aerospace manufacturing design methods, with industry site visits.

High-Cycle Fatigue Testing, UCF Mechanical and Aerospace Engineering (J. L. Kauffman)

Sep 2022 – Dec 2022

Designed a low-cost rig for fatigue testing of repaired turbomachinery blades; implemented a LabVIEW PID controller to hold target stresses.